Problem Solution Fit Report

Food Survey Analysis using Tableau and Flask

The purpose of this document is to demonstrate how the proposed Food Survey Analysis solution directly addresses the problems identified during the requirement analysis phase. The project converts raw survey responses into interactive dashboards, allowing users to identify trends, compare categories, and make data-driven decisions.

Problem Analysis

Educational institutions often collect food survey responses in CSV or spreadsheet format. While the data contains valuable information about food habits, exercise, GPA, healthy feeling, weight, and cuisine preferences, it is difficult to interpret without visualization. Manual analysis consumes time and often fails to reveal meaningful patterns.

Problem	Impact	Proposed Solution
Raw CSV data is difficult to interpret	Users cannot identify trends quickly	Interactive Tableau dashboard with KPIs and charts
Manual analysis is slow	Time-consuming reporting	Automated visual analytics and filters
No relationship analysis	Patterns remain hidden	Scatter plots, histograms, treemaps, and box plots
Static reports	Limited exploration	Interactive dashboard and Tableau Story
Limited accessibility	Users cannot access insights easily	Flask web application with browser access

Why the Proposed Solution Fits

Interactive Dashboard

The dashboard summarizes complex survey data using KPI cards, pie charts, histograms, scatter plots, treemaps, heat maps, and box plots.

User-Centered Design

Filters, labels, tooltips, and dashboards help students, faculty, and researchers explore information without technical knowledge.

Technology Selection

Tableau provides rich visual analytics while Flask enables easy web deployment. GitHub supports version control and documentation.

Scalability

Additional survey records and visualizations can be added without redesigning the application.

Expected Outcomes

The proposed solution improves data understanding, reduces manual effort, highlights relationships among survey variables, and provides an intuitive platform for exploring student food habits and lifestyle patterns.

Evaluation Criterion	Result
Problem Coverage	High
Ease of Use	High
Visualization Quality	High
Scalability	High
Overall Fit	Excellent

Conclusion

The Food Survey Analysis solution is well aligned with the identified problem. By combining Tableau dashboards with a Flask application, the project delivers an efficient, interactive, and scalable analytical platform that meets user needs and project objectives.